

Milestone 1 Technical Understanding

TTP: A Time-Triggered Protocol for Fault-Tolerant Real-Time Systems

Kopetz & Grünsteidl, FTCS 1993

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1 Problem Statement and Motivation

Application domain

- Safety-critical distributed real-time systems (automotive focus)
- Late response = wrong response
- Must tolerate node and channel failures without stopping

Problem with ET systems (e.g. CAN)

- Bus access decided at runtime = collisions under load
- Behaviour differs run to run

Time Triggered

- Full schedule fixed at design time
- Every run identical
- TTP implements TT + fault tolerance + all required services in one protocol

2 Core Technical Contribution

What TTP provides

- Single integrated protocol no separate layers needed
- Services: message transport, clock synchronization, membership, fault tolerance, mode changes, black-out handling

C-state

- Every node maintains C-state: mode + time + membership
- C-state gives all nodes a consistent view of the system

CRC trick

- If receiver C-state differs from sender $\Rightarrow$ CRC fails $\Rightarrow$ frame discarded automatically
- State agreement enforced through error detection

TDMA bus access

- Time divided into fixed slots, one slot per FTU
- One complete cycle through all slots = TDMA round (≈ 1.92 ms)

3 Central Architecture and Formalism

FTU Fault Tolerant Unit

- Group of 1-2 active nodes + optional shadow representing one logical unit
- Both nodes run identical computation, share same TDMA slot
- If active node fails: shadow detects 2 missed rounds, takes over slot

Sparse time base

- Time modelled as discrete grid of points
- All events snap to nearest grid point nothing happens between points
- Small clock differences between nodes tolerated within one granule

4 Assumptions and Scope Limitations

Failure model assumptions

- Nodes are fail-silent: either correct output or no output, never garbage
- Channel failures are omission-only: messages arrive intact or not at all
- Failure rates assumed from Table 1 (e.g. permanent node failure $\approx 10^3/h$)

Schedule is fixed scope limitations

- Schedule generated offline at design time, cannot change at runtime
- Cannot easily add or remove nodes dynamically

5 Runtime, Memory, and Scalability

Timing

- Worst-case message delay = one full TDMA round
- On 250 kbit/s channel: ≈ 1.92 ms (Class I/II), doubled for Class III/IV

Bandwidth efficiency

- TTP overhead: 28 bits per frame vs 47 bits (CAN)
- Data efficiency: 53.3% (TTP) vs 25.3% (CAN)

Scalability

- More nodes = longer TDMA round = higher worst-case latency